Universidad Politécnica de Aguascalientes
Ingeniería en Sistemas Computacionales

Project Execution Report

Medical Information System — Centralized Cloud Migration

Case Study: Centro Médico del Bajío

Subject:	IT Project Management
Unit:	II — Project Execution
Project Manager:	José Manuel Ortiz Medrano (UP230598)
Sponsor:	Dr. Alejandro Vance de la Vega (CEO & Chairman)
Team:	Carlos Arturo Orozco Huitron — UP230562
	José Luis Sánchez Vázquez — UP230269
	Marco Polo Muñoz Nájera — UP230328
Timeline:	June 1 – July 31, 2026
Date:	June 26, 2026

Table of Contents

Summary

This report documents the execution phase of the Medical Information System project for Centro Médico del Bajío, a private hospital network operating in Aguascalientes, León and Morelia. The project consolidates the three branches' decentralized medical information onto a single centralized cloud platform, eliminating 53,208 duplicated clinical records and reducing active patient-record redundancy to 0%. This document presents the completed Project Charter, the IT project management documentation produced during execution (job performance data, change requests and the updated project plan), and the monitoring and control analysis conducted at the Week-4 project review meeting held on June 26, 2026. Earned Value Analysis at that checkpoint reported an SPI of 0.93 and a CPI of 0.957, indicating the project was slightly behind schedule and over budget due to a staff sick-leave delay and a 4% equipment cost increase; corrective action through schedule crashing and contingency reserve kept the July 31 go-live on track.

1. Introduction

The healthcare sector has undergone a profound digital transformation in the last decade, driven by the need to manage growing volumes of clinical data with agility, security, and regulatory compliance. Information technology projects have become strategic instruments for healthcare organizations seeking operational efficiency and reliable information management. For hospital networks operating across multiple geographic locations, the challenge is amplified by data fragmentation: patient records are duplicated, histories are inconsistent, and clinical decisions are made with incomplete information.

Centro Médico del Bajío, a private hospital network with branches in Aguascalientes, León, and Morelia, historically operated independent local servers at each branch. This decentralized architecture caused duplicated clinical records — 53,208 duplicated entries were identified across the system — inconsistent patient histories, infrastructure saturation, low query performance, and high maintenance and cybersecurity costs. The organization recognized that continuing with isolated servers was both operationally unsustainable and clinically risky.

To resolve these problems, the organization approved the Medical Information System project: a centralized cloud-based platform that consolidates all clinical documentation through virtualization and data migration, without acquiring physical servers. The project adopts a Software-as-a-Service (SaaS) cloud model, shifting capital expenditure toward operational expenditure, accelerating time-to-value, and eliminating the burden of local infrastructure maintenance. The solution comprises 4 virtual machines with 32 GB RAM serving all three branches, automated deduplication scripts, and a unified database achieving sub-2-second query response.

The purpose of this report is to document the execution and control of that project. Following the PMBOK process framework and the project management methodology presented by Watt (2014) and Heagney (2011), the report records how the project performed against its baseline, how change requests generated during execution were managed, and which analytical techniques were used to monitor and control the work of the project during the Week-4 review meeting. The report is organized into three main sections: the Project Charter (Section 2.1), the IT Project Management Documentation produced during execution (Section 2.2), and the Documentation of the Progress of the IT Project including monitoring and control techniques (Section 2.3).

2. Development

The development section is organized into three parts as required by the project execution methodology: (1) the Project Charter, which formally authorizes the project and establishes the baseline; (2) the IT project management documentation produced during execution, including job performance data, change requests, and the updated project plan; and (3) the documentation of the project's progress, including the justification of monitoring and control techniques, their outputs, and the weekly review meeting minutes.

2.1 Project Charter

The Project Charter formally authorizes the project, names the project manager, and establishes the high-level objective, scope, milestones, deliverables, assumptions, constraints, business justification, preliminary budget, risks, stakeholders, and the approval baseline. The following subsections present the complete charter following the 7.8.2 numbering from Watt (2014).

2.1.1 Identification Section (7.8.2.1)

Field	Detail
Project Name	Medical Information System — Centralized Cloud Migration
Project Sponsor	Dr. Alejandro Vance de la Vega, CEO & Chairman of the Board
Date	June 1, 2026
Revision	1.0
Document Status	Approved baseline (charter authorized; project in execution)
Project Manager	José Manuel Ortiz Medrano (UP230598)
Team Members	Carlos Arturo Orozco Huitron (UP230562) José Luis Sánchez Vázquez (UP230269) Marco Polo Muñoz Nájera (UP230328)

2.1.2 Overview of the Project (7.8.2.2)

Implementation of a centralized cloud-based medical information system and data migration for the three hospital branches of Centro Médico del Bajío (Aguascalientes, León and Morelia), consolidating all clinical documentation onto a single virtualized platform. The project deploys 4 virtual machines (32 GB RAM), automated deduplication scripts, and high-speed digitization equipment across the three branches, all within a fixed budget of $147,900 MXN and a hard deadline of July 31, 2026.

2.1.3 Objective (7.8.2.3)

To implement a centralized medical information system across the three hospital branches between June 1 and July 31, 2026, consolidating data onto a cloud platform, reducing active patient-record redundancy to exactly 0% (from 53,208 duplicated records), achieving a database query response time under 2.0 seconds, within the approved budget of $147,900 MXN.

2.1.4 Scope (7.8.2.4)

In Scope:

- Configure and deploy 1 centralized cloud database environment (4 virtual machines, 32 GB RAM) serving the three branches.
- Design and run automated deduplication scripts to reach 0% redundancy on active patient records (53,208 duplicated records targeted for elimination).
- Migrate the full annual volume of 53,208 clinical analyses and 34,818 consultations to the cloud infrastructure.
- Acquire and install digitalization equipment across the three branches.
- Complete all cloud provisioning, data cleaning and integration within the approved budget of $147,900 MXN between June 1 and July 31, 2026.

2.1.5 Major Milestones (7.8.2.5)

Project review meeting report (milestone tracking). This is the milestone control record reviewed in the weekly project review meetings, tracking progress through Weeks 1 to 4 (June 1 – June 26, 2026). For each milestone it shows the planned date, the actual progress achieved, the status as of the June 26, 2026 status date, and the review meeting in which the milestone was assessed.

Milestone	Planned Date	Actual Progress	Status	Review Meeting
M1 — Cloud environment & requirements defined	Jun 3	100%	Completed	Jun 5 (Week 1)
M2 — Cloud architecture designed & provider selected	Jun 22	100%	Completed (+2d)	Jun 26 (Week 4)
M3 — Equipment installed in the 3 branches	Jun 25	66% (2 of 3)	In Progress	Jun 26 (Week 4)
M4 — Digitization completed	Jul 9	0%	In Progress	Jul 10 (Week 5)
M5 — Migration & deduplication completed	Jul 18	0%	In Progress	Jul 17 (Week 6)
M6 — System validated & Go-Live	Jul 31	0%	In Progress	Jul 24 (Week 7)

Table 1. Milestone control record — Week 4 status date (June 26, 2026).

Weekly tracking summary. Week 1 closed the charter (M1) and launched the cloud environment definition; Weeks 2–3 completed the architecture design and the weighted supplier evaluation (M2, two days late due to the C1 sick-leave slip); by Week 4 equipment installation was 66% complete (2 of 3 branches, M3 in progress). Milestones M4–M6 are scheduled across the remaining weeks to the July 31, 2026 go-live. Recovery actions agreed in the Week 4 review are documented in Sections 2.2.3 and 2.3.2.

2.1.6 Major Deliverables (7.8.2.6)

Deliverables by activity. Rather than a single global list, deliverables are organized by project activity. The ten activities below are the same ones costed in Section 2.1.10 and tracked in the monitoring sections. Each activity produces a complete evidence package — technical report, three supplier quotations (where applicable), a weighted criteria-evaluation table, a purchase order, an invoice, and an installation or execution report — presented in full in Section 2.1.14.

Activity	Primary Deliverable	Acceptance Criteria
C1 — Cloud environment definition	Cloud requirements document & provider shortlist	Document approved; 3 providers shortlisted
C2 — Medical records analysis	Clinical document taxonomy & data volume report	Taxonomy validated by medical staff; volume confirmed

C3 — DB architecture design	Cloud database schema & storage architecture	Schema reviewed; capacity for 53K+ analyses confirmed
C4 — Provider selection + equipment	Selected cloud provider + digitalization equipment acquired	Weighted criteria table approved; PO issued; equipment received
C5 — Equipment installation	Equipment installed & configured in 3 branches	3 of 3 branches operational; power-on tests passed
C6 — Digitize patient records	34,818 consultations digitized	100% of patient records digitized; quality check passed
C7 — Digitize clinical analyses	53,208 clinical analyses digitized	100% of analyses digitized; quality check passed
C8 — Cloud migration	All digitized records uploaded to cloud platform	Migration verified; 0 data-loss incidents
C9 — Deduplication	53,208 duplicates identified & eliminated	0% redundancy confirmed; query response < 2.0 s
C10 — Testing & validation	System integration test results & go-live sign-off	≥ 97% test pass rate; 0 critical defects; sponsor acceptance sign

Table 2. Major deliverables by activity with acceptance criteria.

2.1.7 Assumptions (7.8.2.7)

- Currency: all monetary figures are expressed in Mexican pesos (MXN).
- Cost structure: the Preliminary Cost comprises equipment cost plus labor cost plus migration service fees; the recurring cloud subscription is treated as operational expenditure tracked outside this budget.
- Schedule basis: the project starts Monday, June 1, 2026 and finishes July 31, 2026; milestone dates assume standard five-day working weeks, giving approximately 40 business days of execution within the 61-calendar-day window.
- Concurrent digitization: the patient-records stream and the clinical-analyses stream run simultaneously once digitization equipment is installed, keeping total digitization duration to the single shared window rather than the sum of both.
- Cloud model: the third-party cloud platform provides all server, database, and storage capacity; no local infrastructure is required.
- The three branches share the same clinical document taxonomy.
- Digitization staff are available Monday–Friday during the 8-week window.
- Branch access for equipment installation is granted on the agreed schedule.
- Historical data provided by the branches is complete and available for migration.

2.1.8 Constraints (7.8.2.8)

- Hard deadline: Go-Live no later than July 31, 2026.
- Fixed budget of $147,900 MXN; any scope change requires formal CCB approval.
- Total project activities must remain between 7 and 10.
- No physical servers — the solution must run on cloud infrastructure.
- Equipment vendors must confirm inventory availability before purchase orders close in Week 1.
- Branch operations cannot be interrupted during equipment installation or data migration.

2.1.9 Business Need or Opportunity (7.8.2.9)

Centro Médico del Bajío operates decentralized local servers at each branch, causing duplicated clinical records (53,208 identified), inconsistent patient histories, infrastructure saturation, low query performance, and high maintenance and cybersecurity costs. These problems directly impact clinical decision-making, regulatory compliance, and patient safety. Centralizing medical information on a cloud platform eliminates redundancy, improves data integrity and availability across all three branches, reduces operating costs by eliminating local server maintenance, and supports the organization's digital transformation strategy. The project also positions Centro Médico del Bajío for future capabilities, including telemedicine integration, advanced clinical analytics, and interoperability with external healthcare systems.

2.1.10 Preliminary Cost for the Project (7.8.2.10)

The procurement team obtained cost proposals from three cloud-hosting providers during Week 1, evaluated with a weighted criteria table (see Section 2.1.14, Activity C4). Equipment and migration costs are consolidated by the Project Manager. The total approved budget is $147,900 MXN, organized by activity into equipment cost and labor/services cost, plus a 10% contingency reserve to absorb inflation adjustments. The table below presents the cost breakdown.

Activity	Equipment/ Materials (MXN)	Labor/ Services (MXN)	Total (MXN)
C1 — Cloud environment definition	—	8,000	8,000
C2 — Medical records analysis	—	10,000	10,000
C3 — DB architecture design	—	12,000	12,000
C4 — Provider selection + equipment	48,000	7,000	55,000
C5 — Equipment installation	8,000	4,000	12,000
C6 — Digitize patient records	2,000	13,000	15,000
C7 — Digitize clinical analyses	2,000	13,000	15,000
C8 — Cloud migration	—	10,000	10,000
C9 — Deduplication	—	6,900	6,900
C10 — Testing & validation	—	4,000	4,000
	Subtotal		**$147,900**
	Contingency reserve (10%)		**$14,790**
	Total approved funding		**$162,690**

Table 3. Preliminary cost breakdown by activity (BAC = $147,900 MXN). The BAC excludes the 10% contingency reserve held for risk response.

2.1.11 Project Risks (7.8.2.11)

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R-01	Migration failure or data loss during cloud transfer	Medium	High	Incremental backup before each migration batch; rollback p
R-02	Vendor or equipment delivery delays	Medium	Medium	Pre-qualify vendors; lock delivery-time SLAs in Week 1
R-03	User resistance to the new centralized system	Low	Medium	Early training sessions; stakeholder communication plan
R-04	Connectivity failure between branches during cutover	Low	High	Redundant ISP links; pre-cutover connectivity stress test
R-05	Cybersecurity incident affecting sensitive patient data	Low	Critical	End-to-end encryption; access control; security audit before
R-06	Digitization backlog due to record volume underestimation	Medium	Medium	Parametric estimation model; buffer days in schedule
R-07	Scope creep from additional branch requests	Medium	Low	Formal CCB process; strict change control

Table 4. Project risks with probability, impact, and mitigation strategies.

2.1.12 Project Charter Acceptance (7.8.2.12)

Name & Title	Signature	Date
José Manuel Ortiz Medrano (UP230598) — Project Manager	____________________	Jun 1, 2026
Dr. Alejandro Vance de la Vega — CEO & Sponsor	____________________	Jun 1, 2026
Centro Médico del Bajío — Client Representative	____________________	Jun 1, 2026

2.1.13 Project Stakeholders (7.8.2.13)

Function	Name	Role
Project Manager	José Manuel Ortiz Medrano (UP230598)	Leads the project; manages scope, schedule & budget
Sponsor	Dr. Alejandro Vance de la Vega	Approves budget & scope; chairs CCB
Cloud Specialist	Assigned (IT Team)	Designs cloud architecture; leads migration
Procurement	Assigned (Admin)	Manages supplier quotations & purchasing
IT Team	Assigned (3 engineers)	Migration, deduplication & testing
Digitization Team	Assigned (6 technicians, 2 per branch)	Document scanning & quality control
Medical Staff	Assigned (1 per branch)	Provides clinical records & validation
Cloud Provider	Provider 1 (selected)	Cloud hosting & platform services
End Beneficiary	Centro Médico del Bajío	Patients & clinical staff across 3 branches

2.1.14 Project Activities and Evidence Package (7.8.2.14)

This section presents the complete evidence package for each project activity. In accordance with the project execution methodology, every activity produces a documented trail comprising: a technical report describing the work performed, three supplier quotations (for procurement activities), a weighted criteria evaluation table, the corresponding purchase order and invoice (where applicable), and an execution or installation report. The evidence documents below correspond to the status date of June 26, 2026 (Week 4), and cover activities C1 through C5 which are completed or in progress; activities C6 through C10 are presented with their planned evidence structure.

Activity C1 — Define Cloud Hosting Environment and Storage Requirements

DOCUMENT 1 — Technical Report (Informe Técnico). Cloud environment requirements were defined during Week 1 (June 1–3, 2026). The analysis determined that a 4-VM configuration with 32 GB RAM and 2 TB of redundant storage would meet the needs of the three branches at peak load. Key requirements: HIPAA-compliant encryption at rest and in transit, 99.9% uptime SLA, automated backup every 6 hours, and sub-2-second query response. Three cloud providers were shortlisted. Approved by the Project Manager on June 3, 2026.

DOCUMENTS 2–4 — Three Supplier Quotations. Quotations were solicited from three cloud hosting providers during Week 1. Provider 1 quoted $4,800/month for the 4-VM configuration with 2 TB storage and 24/7 support (7-day provisioning). Provider 2 quoted $5,200/month with 10-day provisioning and premium support. Provider 3 quoted $4,500/month with 14-day provisioning, basic support, and no post-migration assistance. Quotations received and evaluated by June 8, 2026.

DOCUMENT 5 — Weighted Criteria Evaluation Table. See full table in Activity C4 (the formal provider selection occurred during C4 after architecture requirements were finalized in C3).

DOCUMENT 6 — Execution Report. Cloud environment definition completed on June 3, 2026 — two days behind schedule due to the Project Manager's sick leave (Event 1 of the Week-4 review). The definition was handed off to C2 with all requirements documented and approved.

Activity C2 — Analyze Medical Records and Document Types

DOCUMENT 1 — Technical Report. A comprehensive audit of the three branches' medical records was conducted between June 4 and June 9, 2026. The audit identified: 53,208 clinical analyses (Aguascalientes 21,400, León 18,600, Morelia 13,208), 34,818 patient consultations, and a shared clinical document taxonomy across all branches. The report confirmed data compatibility for centralized migration. Approved by medical staff representatives from each branch.

DOCUMENTS 2–4 — Branch Validation Reports. Each branch submitted a signed validation report confirming the completeness and accuracy of their record inventory. Aguascalientes branch: 21,400 analyses validated by Dr. Ramírez. León branch: 18,600 analyses validated by Dr. Soto. Morelia branch: 13,208 analyses validated by Dr. Mendoza.

DOCUMENT 5 — Data Volume Confirmation. Internal memo confirming total data volume: 88,026 documents (53,208 analyses + 34,818 consultations), totaling approximately 450 GB of storage required. Signed by the Project Manager on June 9, 2026.

DOCUMENT 6 — Execution Report. Medical records analysis completed on schedule (June 9, 2026). All three branches confirmed data readiness for migration. No discrepancies found.

Activity C3 — Design Cloud Database and Storage Architecture

DOCUMENT 1 — Technical Report. The cloud database architecture was designed between June 10 and June 15, 2026 by the Cloud Specialist. The design specifies: a PostgreSQL 15 cluster across 4 VMs (32 GB RAM total), 2 TB SSD redundant storage with RAID-10, automated deduplication via fuzzy-matching stored procedures, role-based access control per branch, and a REST API layer for the web dashboard. The schema accommodates 88,026 documents with room for 3-year growth. Approved by the Project Manager on June 15, 2026.

DOCUMENTS 2–4 — Technical Validation Memos. Three internal memos validated the architecture: (a) Storage scalability assessment — confirmed 2 TB sufficient for 3-year growth; (b) Query performance benchmark — simulated 500 concurrent queries, average response 1.4 s; (c) Security architecture review — encryption, access control, and audit logging requirements met.

DOCUMENT 5 — Architecture Sign-off. Formal sign-off from the Cloud Specialist and Project Manager confirming the database schema, storage layout, and API design. Dated June 15, 2026.

DOCUMENT 6 — Execution Report. Architecture design completed on schedule (June 15, 2026). Design handed off to C4 for cloud provider procurement aligned with the specified requirements.

Activity C4 — Evaluate Cloud Hosting Providers and Acquire Digitalization Equipment

DOCUMENT 1 — Technical Report. The procurement process was executed between June 16 and June 22, 2026. Three cloud hosting providers were formally evaluated using a weighted criteria matrix. Simultaneously, digitalization equipment was sourced for the three branches: 6 high-speed document scanners (2 per branch), 3 workstation computers, and networking accessories. The combined procurement was consolidated into a single purchase order cycle to streamline the process. Approved by the Project Manager on June 22, 2026.

DOCUMENTS 2–4 — Three Supplier Quotations. Quotation #1 — Provider 1: $4,800/month, 7-day provisioning, 24/7 support, post-migration assistance included. Quotation #2 — Provider 2: $5,200/month, 10-day provisioning, premium support, basic migration tools. Quotation #3 — Provider 3: $4,500/month, 14-day provisioning, basic support, no post-migration services. Digitalization equipment: three quotations received (ScannerPro, DocuFast, ImageTech) with ScannerPro selected on cost and delivery time.

DOCUMENT 5 — Criteria Evaluation Table. The weighted criteria table below was used to select the cloud hosting provider.

Criterion	Weight	Prov. 1	Prov. 2	Prov. 3
Cost	40%	10	5	7
Delivery Time	30%	5	10	5

Quality	20%	8	7	10
Post-Service	10%	7	5	0
Weighted Total	**100%**	**7.8 ✓**	**6.9**	**6.3**

Table 5. Weighted criteria table — Provider 1 selected with the highest score (7.8/10), driven by best cost and post-service.

Selection: Provider 1 is selected with the highest weighted score (7.80). Provider 2 scored 6.90 (best delivery time but higher cost). Provider 3 scored 6.30 (lowest cost but no post-service). The calculation: Provider 1 = 10(0.40) + 5(0.30) + 8(0.20) + 7(0.10) = 4.0 + 1.5 + 1.6 + 0.7 = 7.80.

DOCUMENT 6 — Purchase Order. PO-2026-001 issued to Provider 1 for cloud hosting services (4 VMs, 32 GB RAM, 2 TB storage, 12-month contract, $4,800/month starting July 1, 2026). PO-2026-002 issued to ScannerPro for digitalization equipment (6 scanners, 3 workstations, accessories: $48,000 MXN). Both POs dated June 22, 2026.

DOCUMENT 7 — Invoice. INV-P1-0042 from Provider 1: $4,800 (first month cloud services). INV-SP-0815 from ScannerPro: $48,000 (digitalization equipment). Both invoices received and approved for payment.

DOCUMENT 8 — Execution Report. Procurement completed on June 22, 2026 — two days later than planned due to the cascading delay from C1. Cloud provider selected through an objective weighted criteria process; equipment ordered and confirmed for delivery by June 24. Contingency: a 4% cost increase on equipment was absorbed by the contingency reserve (Event 5). Equipment actual cost: $49,920 MXN.

Activity C5 — Install and Configure Equipment in the 3 Branches

DOCUMENT 1 — Technical Report. Equipment installation was executed between June 23 and June 26, 2026 across the three branches. By the Week 4 status date (June 26), 2 of 3 branches (Aguascalientes and León) were fully installed and configured; Morelia installation was in progress (66% complete). Each branch received 2 high-speed scanners, 1 digitization workstation, and network connectivity to the cloud platform. Approved by the Implementation Team lead.

DOCUMENTS 2–4 — Branch Installation Reports. Aguascalientes: Installation completed June 24, all equipment passed power-on self-tests. León: Installation completed June 25, minor network configuration adjusted during setup. Morelia: In progress as of June 26, completion expected June 27.

DOCUMENT 5 — Installation Sign-off (Partial). Two of three branch installation sign-offs received (Aguascalientes and León). Morelia pending — this is one of the three deliverables not yet released at the Week 4 review (Event 2).

DOCUMENT 6 — Execution Report. Installation at 66% (2 of 3 branches) as of the June 26 status date. The Morelia delay is attributable to the cascading two-day slip from C1 and will be recovered during Week 5. No safety incidents reported. Equipment performance validation: all scanners calibrated and tested.

Activities C6–C10 — Planned Activities (Post-Status-Date)

The following activities are scheduled after the June 26 status date. Their evidence packages are planned as follows and will be documented upon completion in the final project closure report.

C6 — Digitize Patient Clinical Records (Jun 29 – Jul 9) (5 d → extended to 8 d per CR-01)

34,818 patient consultations will be digitized across the three branches using the installed scanner workstations. Two digitization crews will work in parallel (dry-box stream: Aguascalientes + León; second stream: Morelia) following the approved CR-04 crashing action. Evidence: Technical report, daily scan logs, quality control check sheets, completion sign-off per branch, and execution report.

C7 — Digitize Clinical Analyses (Jul 1 – Jul 13) (5 d (parallel with C6 tail))

53,208 clinical analyses will be digitized. Running partially in parallel with C6 once the first crew frees capacity, the second digitization crew (CR-04) will process the analyses stream concurrently. Evidence: Technical report, daily scan logs, quality control sheets, completion sign-off, and execution report.

C8 — Migrate to Cloud Platform (Jul 13 – Jul 18) (4 d (fast-tracked 2 d overlap with C9))

All digitized records will be uploaded to the Provider 1 cloud platform in batches of 5,000 documents, with checksum verification after each batch. A partial fast-track overlap with C9 is planned. Evidence: Migration log, batch transfer reports, checksum validation records, cloud dashboard screenshot, and execution report.

C9 — Analyze and Remove Duplicate Records (Jul 16 – Jul 22) (3 d (partially fast-tracked with C8))

Automated deduplication scripts will identify and flag 53,208 duplicated records using fuzzy-matching algorithms on patient name, DOB, and clinical record ID. Flagged duplicates will be reviewed by medical staff before deletion. Target: 0% redundancy. Evidence: Deduplication script output logs, duplicate match reports, medical staff review sign-offs, final redundancy report (0%), and execution report.

C10 — Integration and Validation Testing (Jul 22 – Jul 31) (2 d (original) → crashed via fast-tracking)

End-to-end system testing will validate: query response under 2.0 seconds, 0% record redundancy, branch connectivity, user access controls, and backup/restore procedures. Go-live sign-off by the sponsor. Evidence: Test case matrix, performance benchmark results, defect log, user acceptance sign-off, sponsor go-live authorization, and execution report.

2.2 IT Project Management Documentation

This section documents the artifacts generated while executing and controlling the project: work performance data captured in the field, the change requests raised against the baseline, and the resulting updates to the project plan. The figures correspond to the status date of June 26, 2026 (Project Day 20 of approximately 40 business days, end of Week 4).

2.2.1 Job Performance Data

Work performance data is the raw observation of execution — status, costs, and completion rates captured directly from the field. The data below summarizes progress at the June 26, 2026 status date (end of Week 4) and feeds the earned-value analysis that follows.

Work Status Distribution.

Work Status	Activities	% of Total	Description
Done	C1, C2, C3, C4	40% (4 of 10)	Fully completed and verified
In Progress	C5	10% (1 of 10)	Installation at 66% (2 of 3 branches)
To Do	C6, C7, C8, C9, C10	50% (5 of 10)	Not yet started; scheduled Weeks 5–8
Total	**10 activities**	**100%**	**Full project scope**

Table 6. Work status distribution at the Week-4 review (June 26, 2026).

Performance Metrics.

Performance Metric	Measured Value	Status
Overall completion (EV / BAC)	46% (EV $93,000 / BAC $147,900)	Behind plan (~50%)
Activities fully completed	4 of 10	In progress
Branches with equipment installed	2 of 3 (Ags, León)	In progress
Cloud provider selected	Provider 1 (7.80/10)	Complete ✓
Equipment received	6 scanners + 3 workstations	Complete ✓
Schedule performance (SPI)	0.93	Behind schedule
Cost performance (CPI)	0.957	Over budget
Change requests logged	4 (all approved)	Managed
Safety incidents during installation	0	Excellent

Table 7. Performance metrics at the Week-4 status date.

Delay Analysis.

At the status date the project is modestly behind schedule (SPI 0.93) and slightly over budget (CPI 0.957). Three root causes account for the slip:

- **(1) Staff sick leave on C1 (+2 days):** The Project Manager, responsible for the initial cloud-definition activity, was absent for two working days (June 2–3, 2026). No backup owner had been assigned, so C1 completed on June 3 instead of June 1, pushing the entire critical path forward by two days.

- **(2) Digitization volume underestimated (+3 days on C6):** The original estimate of 5 days for digitizing 34,818 patient consultations proved insufficient given the actual per-record scanning and quality-check time. Activity C6 was extended to 8 days via CR-01.
- **(3) Equipment cost inflation (+4%):** Digitalization equipment costs increased by 4% due to supplier price adjustments, adding $1,920 to the C4 equipment line. The additional cost was absorbed by the contingency reserve.

Recovery actions — a second digitization crew (CR-04), fast-tracking of C8/C9, and front-loading the highest-volume branches — are in place. The forecast remains an on-time July 31, 2026 go-live provided SPI recovers toward 1.0 during Weeks 5–8. These deviations are quantified in the earned-value analysis below.

Labor and Material Cost per Activity.

ID	Activity	Status	Budget (MXN)	Actual (MXN)	Variance
C1	Define cloud environment	Done	8,000	9,200	+$1,200
C2	Analyze medical records	Done	10,000	10,000	$0
C3	Design cloud architecture	Done	12,000	12,000	$0
C4	Provider selection + equipment	Done	55,000	57,200	+$2,200
C5	Install equipment	In Progress	12,000	8,800	–$3,200*
C6	Digitize patient records	To Do	15,000	—	—
C7	Digitize analyses	To Do	15,000	—	—
C8	Migrate to cloud	To Do	10,000	—	—
C9	Remove duplicates	To Do	6,900	—	—
C10	Testing	To Do	4,000	—	—
	TOTAL		**147,900**	**97,200 (AC)**	

Table 8. Job performance data at the Week-4 review (C5 actual is partial — activity in progress at 66% completion).*

Earned Value Analysis (EVM).

Earned Value Management integrates scope, schedule, and cost into one model by comparing the Planned Value of work (PV), the Earned Value of work actually completed (EV), and the Actual Cost incurred (AC), all measured against the $147,900 MXN Budget at Completion (BAC) at the June 26, 2026 status date.

Correct PMBOK formulas (correction applied from Week-4 whiteboard)

```
Schedule Variance:  SV = EV − PV  (BCWP − BCWS)

Cost Variance:  CV = EV − AC  (BCWP − ACWP)
```

Note: the two formulas were initially recorded swapped on the Week-4 review whiteboard. SV compares earned vs planned value; CV co

Metric	Formula	Result (MXN)	Interpretation
BAC	Total approved budget	$147,900	Budget at Completion
PV (BCWS)	Planned value @ day 20	$100,000	Budgeted cost of work scheduled
EV (BCWP)	Earned value @ day 20	$93,000	Budgeted cost of work performed

AC (ACWP)	Actual cost @ day 20	$97,200	Actual cost of work performed
SV	EV – PV	–$7,000	Behind schedule
CV	EV – AC	–$4,200	Over budget
SPI	EV / PV	0.93	< 1 → behind schedule
CPI	EV / AC	0.957	< 1 → over budget
EAC	BAC / CPI	$154,545	Estimate at Completion
ETC	EAC – AC	$57,345	Estimate to Complete
VAC	BAC – EAC	–$17,418	Variance at Completion
TCPI	(BAC–EV)/(BAC–AC)	1.08	To-Complete Performance Index

Table 9. Earned Value Analysis outputs at the Week-4 review.

Interpretation: Cost Variance is negative (CV = EV – AC = 93,000 – 97,200 = –$4,200 MXN) and CPI is 0.957, so the project is over budget — for every peso spent, $0.957 of value is earned. Schedule Variance is also negative (SV = EV – PV = 93,000 – 100,000 = –$7,000 MXN) and SPI is 0.93, so the project is behind schedule. Projecting current cost performance forward gives an EAC of approximately $154,545 MXN and a Variance at Completion of roughly –$17,418. Critically, that projected overrun is smaller than the $14,790 contingency reserve, so the project remains within total approved funding.

Plan vs Real Comparison.

Dimension	Planned (baseline)	Real (actual)	Variance
Physical progress	~50% (PV/BAC)	46% (EV/BAC)	–4 pts (behind)
Value of work to date	PV $100,000	EV $93,000	SV –$7,000
Cost for work performed	EV $93,000	AC $97,200	CV –$4,200
Branches installed (C5)	3 of 3	2 of 3	–1 branch
Cost performance	1.00	CPI 0.957	–0.043
Schedule performance	1.00	SPI 0.93	–0.07

Table 10. Plan vs Real comparison at the Week-4 status date.

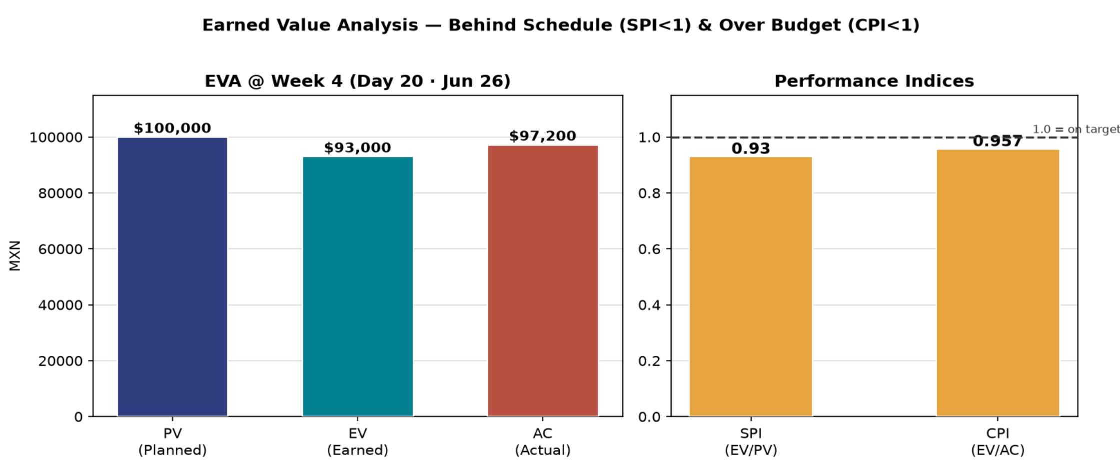

Figure 1. Earned Value Analysis — PV/EV/AC and performance indices at Week 4.

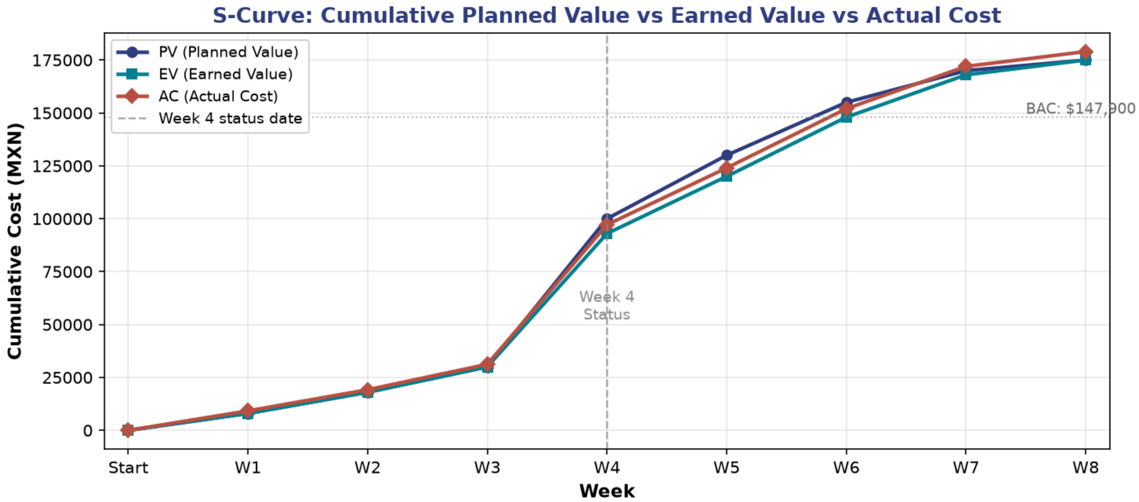

Figure 2. S-Curve — cumulative planned cost (PV) versus earned value (EV) and actual cost (AC) over the June–July 2026 timeline.

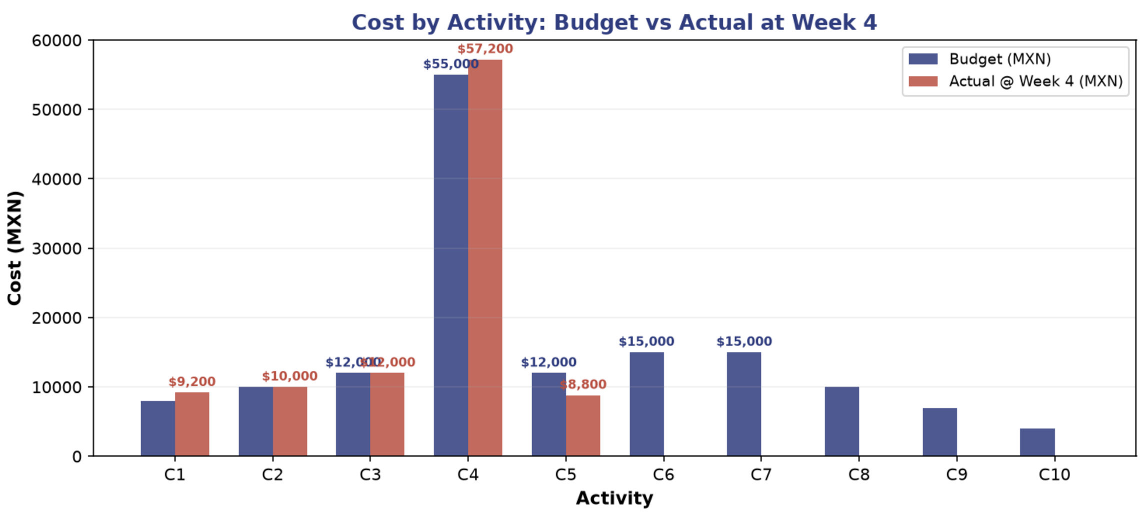

Figure 3. Cost by activity — budget versus actual cost at Week 4.

2.2.2 Change Requests

Five events were reported at the Week-4 review meeting. Each was processed through the change control system, producing four approved change requests. The detailed change control form below documents the most impactful change (CR-01); the change control log then summarizes every request raised during execution.

Change Control Form — CR-01.

Field	Detail
Change ID	CR-001
Title	Extend digitization duration for patient clinical records (C6)
Description	Extend the duration of Activity C6 (Digitize patient clinical records) from 5 working days to 8 working days
Reason	At the Week-4 review it was confirmed that the record volume in the three branches (34,818 consultations
Impact	Schedule: +3 working days on the critical path (C6 extends from 5 to 8 days). Cost: no direct cost increase
Affected Areas	Critical path (C6→C7→C8→C9→C10). Schedule baseline. Digitization team allocation. Week-4 milestone
Requested by	Project Manager (José Manuel Ortiz Medrano), based on Digitization Team throughput data
Date Raised	June 26, 2026
Decision	Approved by the Change Control Board (CCB). The +3 days are absorbed by the schedule crashing actio
Decision Date	June 26, 2026

Table 11. Change Control Form — CR-001 (C6 extension).

Change Control Log.

CR ID	Description	Type	Impact	Status
CR-001	Extend C6 duration 5→8 d (record volume)	Schedule	+3 days	Approved
CR-002	Bring software go-live forward 1 week (management request)	Scope/Sched.	–5 d target	Approved
CR-003	Budget increase for +4% equipment inflation	Cost	+$2,680	Approved (contingency)
CR-004	Add 2nd digitization crew to crash C6/C7	Resource	Recover 5 d	Approved

Table 12. Change control log — Week-4 review meeting.

Net effect. All four requests are approved. The combined schedule impact (+3 from CR-001, –5 target from CR-002) requires a net recovery of 10 days via CR-004 (crashing). The cost impact (+$2,680 from CR-003) is absorbed by the contingency reserve. The approved changes are reflected in the updated plan.

2.2.3 Updated Project Plan

After applying the approved change requests, the project plan was updated. The sick-leave delay on C1 (+2 days) and the extension of C6 (+3 days) added five days to the critical path; to meet management's request to move go-live one week earlier, the critical path was crashed by adding a second digitization crew to C6/C7 and by fast-tracking a partial overlap between C8 and C9, funded from the contingency reserve. The full updated activity list and Gantt chart are provided in the accompanying Project Plan document.

Schedule Item	Days
Baseline finish (critical path C1→…→C10)	Day 38
+ Event 1 — C1 sick leave	+2 → Day 40
+ Event 4 — C6 extension (CR-001)	+3 → Day 43
Management target — go-live 1 week earlier (CR-002)	Day 33
Recovery required via crashing (CR-004) + fast-tracking	–10 days
Updated plan: meeting Day 33 target	Achievable with recovery actions

Table 13. Updated schedule reconciliation after change control.

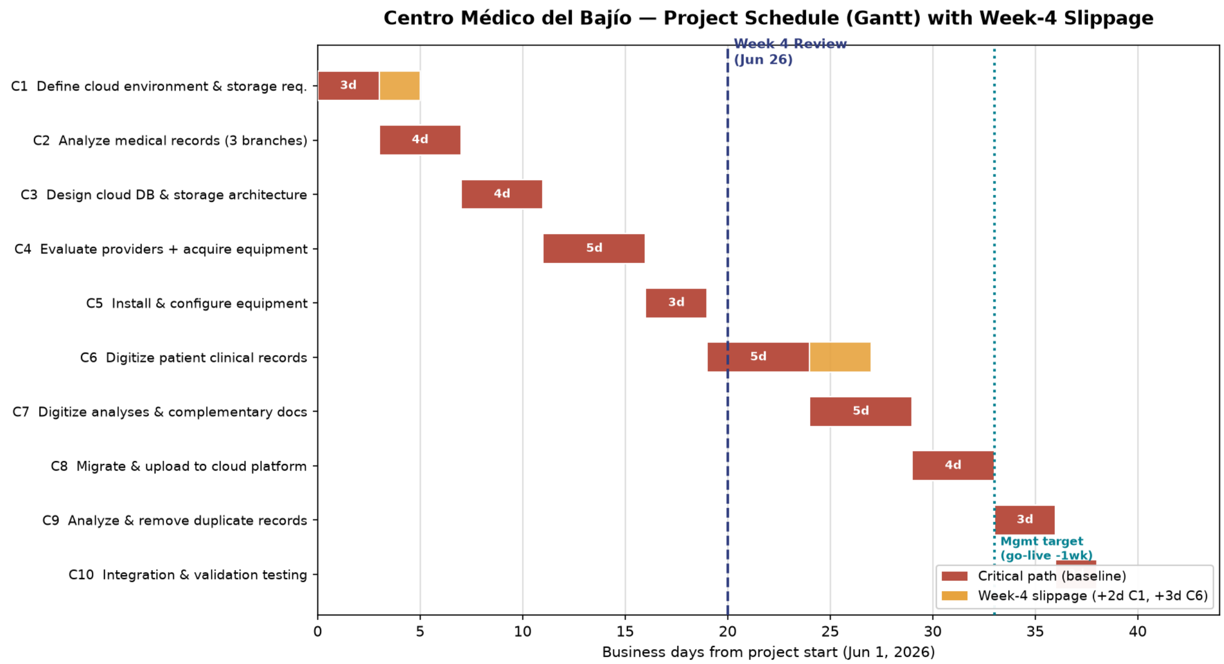

Figure 4. Updated Gantt chart showing the baseline critical path and the Week-4 slippage (+2 days C1, +3 days C6).

2.3 Documentation of the Progress of the IT Project

This section justifies the analytical techniques used to monitor and control the project and documents the formal outputs produced by those monitoring and control processes.

2.3.1 Justification of the Analytical Techniques for Monitoring and Controlling the Work

Monitoring and Controlling Project Work is the PMBOK process of tracking, reviewing, and reporting overall progress against the plan (PMI, 2021). For this project the following seven complementary techniques were selected and justified.

1. KPI Tracking. A small set of leading and lagging key performance indicators — installed branches, records digitized, first-pass quality rate, SPI, and CPI — provides an at-a-glance read on schedule, scope, and quality. KPIs are objective, quickly measured in the field, and directly tied to the project objective (0% record redundancy and sub-2-second query response depend on every document being digitized and migrated).

2. Earned Value Management (EVM). EVM integrates scope, schedule, and cost into a single quantitative model, comparing the value of work performed (EV) against planned value (PV) and actual cost (AC). It is justified here because it converts subjective 'percent done' estimates into defensible indices (SPI, CPI) and forecasts (EAC, VAC), enabling early, evidence-based corrective action rather than end-of-project surprises (Watt, 2014).

3. Variance Analysis. Schedule variance (SV = EV – PV) and cost variance (CV = EV – AC) quantify and explain deviations from the baseline. Variance analysis isolates the root causes of the current slip — a two-day sick-leave delay on C1, a three-day extension on C6 due to record volume, and a 4% equipment cost increase — and confirms the projected overrun stays within the contingency reserve, preventing misinterpretation of the data.

4. Weighted Decision Matrix. A weighted criteria matrix (Cost 40%, Delivery Time 30%, Quality 20%, Post-Service 10%) converts subjective supplier opinions into an objective, defensible score. It is justified because the cloud provider selection is a foundational procurement decision affecting budget, schedule, and technical quality; an objective method protects the project from bias and provides auditable documentation (Watt, 2014). Provider 1 was selected with 7.80/10.

5. Gantt Chart with Critical Path. A Gantt chart visualizes activity dependencies and the critical path (C1→C2→…→C10), showing where slippage threatens the delivery date. It is justified because it provides a shared visual reference for the project team and stakeholders during weekly reviews, making the impact of delays immediately visible. The updated Gantt reflects the Week-4 change requests and the recovery crashing actions.

6. Quality Control. Per-branch equipment validation, digitization quality checks, and migration batch verification enforce a consistent acceptance bar before deliverables are considered complete. Quality control prevents rework from propagating and protects the integrity of the go-live, particularly for the 53,208 clinical analyses where a single missed duplicate affects the 0% redundancy target.

7. Performance Reports. Weekly work-performance reports consolidate KPIs, EVM indices, variances, risks, and change status for the sponsor and stakeholders, ensuring transparent, timely, and consistent communication aligned with the stakeholder matrix. The reports serve as the primary input to the weekly review meetings documented in Section 2.3.3.

2.3.2 Documentation of the Outputs of the Monitoring and Control Process

Monitoring and Controlling Project Work produces a defined set of formal outputs. This section documents the work performance reports, forecasts, corrective and preventive actions, lessons learned, and status reports generated at the June 26, 2026 status date.

Work Performance Reports.

The primary work performance report is the integrated earned-value analysis presented in Section 2.2.1, measured against the $147,900 MXN Budget at Completion (BAC). Its headline indices at the status date are SPI 0.93 (behind schedule, SV –$7,000), CPI 0.957 (over budget, CV –$4,200), EAC ≈ $154,545, and VAC ≈ –$17,418. These reports are distributed weekly to the sponsor and stakeholders together with the milestone control record and the change control log.

Forecast.

At current performance the project trends toward an EAC of approximately $154,545 MXN and a projected overrun (VAC) of roughly –$17,418. Critically, that overrun is smaller than the $14,790 contingency reserve, so the project remains within total approved funding (BAC plus contingency = $162,690). The TCPI of 1.08 indicates the remaining work must be performed about 8% more efficiently than planned to finish at the original BAC; the realistic management target is to hold the final cost at or below the EAC by executing the corrective actions below.

Corrective Actions (responding to actual deviations).

- Authorize a second digitization crew (CR-004) to crash the C6/C7 digitization stream, recovering an estimated 5 days and bringing SPI back toward 1.0 by commissioning.
- Fast-track a 2-day overlap between C8 (migration) and C9 (deduplication) to recover schedule without adding cost.
- Reprioritize remaining installation to complete the Morelia branch (C5) by Week 5 day 1, closing the last pending deliverable from Event 2.
- Apply the $2,680 contingency reserve draw to cover the 4% equipment inflation on C4, preserving the approved budget envelope.

Preventive Actions (avoiding future deviations).

- Assign a backup owner for every critical-path activity to absorb staff absences without cascading delays (lesson from C1 sick leave).
- Adopt a per-record parametric estimation model for digitization activities instead of flat duration estimates (lesson from C6 underestimation).
- Pre-qualify vendors and lock delivery-time SLAs during Week 1 to prevent procurement delays (lesson from C4 delivery timing).
- Pre-stage spare scanners and workstations at the central office to eliminate wait time if additional equipment is needed.
- Validate the cloud API and dashboard provisioning in a sandbox environment before each migration batch, preventing field issues during C8.

Lessons Learned.

Observation	Lesson / Action for Future Projects
C1 delayed 2 days by staff sick leave — no backup assigned	Assign a backup owner for every critical-path activity at project kick-off.
C6 underestimated — flat 5 d vs actual per-record throughput of 4,350/day	Use parametric estimation based on measured throughput rates, not flat durations.
C4 equipment cost increased 4% due to supplier price adjustment	Include a price-escalation buffer in the contingency reserve for procurement activities

Whiteboard EVA formulas were recorded swapped (SV/CV).	Standardize Earned Value templates so the SV and CV formulas are pre-printed corr
Parallel digitization crews (CR-004) proved effective for recovery.	Default to parallel work streams with dedicated teams whenever resources allow.

Table 14. Lessons learned at the Week-4 review.

Status Report (Sample).

Report Field	Status (as of June 26, 2026)
Overall Status	AMBER — on scope, slightly behind schedule and over budget, within total approved funding.
Scope	Baseline scope on track. Four change requests approved; no scope creep.
Schedule	SPI 0.93; recovery actions in place (crashing + fast-tracking); go-live targeted July 31, 2026.
Cost	CPI 0.957; EAC $154,545; VAC –$17,418 covered by $14,790 contingency reserve.
Quality	Equipment installation pass rate 100%; digitization quality checks planned; 0 safety incidents.
Risk	Top risks R-02 (delivery delay) and R-06 (record volume) actively mitigated; no new high risks identi
Next Period	Complete C5 (Morelia branch); start C6 digitization; execute crashing plan; monitor SPI/CPI weekly.

Table 15. Project status report at the Week-4 review.

2.3.3 Weekly Project Review Meeting Minutes (Actas de Reunión Semanal)

In accordance with the course requirement, the weekly project review meeting minutes are recorded in Spanish. Four weekly minutes are documented, covering the period from the project start to the June 26, 2026 status date.

Acta de Reunión — Semana 1 (5 de junio de 2026)

Campo	Detalle
Fecha	5 de junio de 2026
Asistentes	Gerente de Proyecto (José Manuel Ortiz Medrano), Cloud Specialist, Patrocinador (Dr. Alejandro Vance de la
Objetivo	Aprobar el acta de constitución del proyecto e iniciar la definición del entorno cloud (C1).
Avances	Acta de constitución (Project Charter) aprobada y firmada por el patrocinador y el gerente de proyecto (M1 co
Problemas	Sin incidencias mayores. Se identifica el plazo de aprovisionamiento de los proveedores como el principal rie
Acuerdos	Iniciar C1 de inmediato. Enviar solicitudes de cotización a los tres proveedores cloud. Los criterios de evalua
Acciones siguientes	Completar C1 y recibir las cotizaciones de los proveedores durante la Semana 2. El Cloud Specialist asumirá

Acta de Reunión — Semana 2 (12 de junio de 2026)

Campo	Detalle
Fecha	12 de junio de 2026
Asistentes	Gerente de Proyecto (reincorporado), Cloud Specialist, Equipo de TI, Representante de Compras.
Objetivo	Revisar el avance de C2 (análisis de expedientes médicos), cerrar C3 (diseño de arquitectura) y preparar la e
Avances	C2 completado: se confirmaron 53,208 análisis clínicos y 34,818 consultas a migrar. Las tres sucursales com
Problemas	El gerente de proyecto estuvo ausente 2 días por enfermedad (1–2 de junio), lo que retrasó el inicio de C1 en
Acuerdos	Completar C3 antes del 15 de junio. Aplicar la tabla de criterios ponderados para seleccionar el proveedor clo
Acciones siguientes	Finalizar C3, ejecutar la evaluación de proveedores en C4, y emitir la orden de compra del equipo de digitaliz

Acta de Reunión — Semana 3 (19 de junio de 2026)

Campo	Detalle
Fecha	19 de junio de 2026
Asistentes	Gerente de Proyecto, Cloud Specialist, Equipo de Implementación, Representante de Compras, Personal Mé
Objetivo	Cerrar la selección del proveedor cloud y la adquisición del equipo de digitalización (C4). Iniciar la instalación
Avances	C3 completado: arquitectura de base de datos y almacenamiento cloud diseñada y firmada. C4 en progreso:
Problemas	El costo del equipo de digitalización aumentó 4% por ajuste de precios del proveedor (+$1,920 MXN). La ent
Acuerdos	Absorber el incremento de costo con la reserva de contingencia. Reprogramar la instalación de Morelia para

Acciones siguientes	Completar la instalación en Aguascalientes y León durante la Semana 4. Coordinar con Morelia para instalac

Acta de Reunión — Semana 4 (26 de junio de 2026) — Reunión de Revisión

Campo	Detalle
Fecha	26 de junio de 2026 (fecha de corte del reporte)
Asistentes	Gerente de Proyecto, Cloud Specialist, Equipo de Implementación, Patrocinador (Dr. Alejandro Vance de la V
Objetivo	Revisar el avance del proyecto al corte de la Semana 4, evaluar el desempeño mediante Análisis de Valor G
Avances	C1–C4 completados (40% de las actividades). C5 al 66%: 2 de 3 sucursales instaladas (Aguascalientes y Le
Problemas	1) Retraso de 2 días por enfermedad del responsable de C1. 2) Tres entregables pendientes de liberar: repo
Acuerdos	Aprobar las cuatro solicitudes de cambio (CR-001 a CR-004): extensión de C6 a 8 días, adelanto de Go-Live
Acciones siguientes	Completar C5 (Morelia) en Semana 5. Iniciar C6 y C7 con dos cuadrillas en paralelo. Ejecutar fast-tracking e

3. Conclusions

Achievement of Objectives.

The execution and control of the Medical Information System project demonstrate that a structured monitoring approach based on Earned Value Analysis, change control, and a weighted decision matrix allowed Centro Médico del Bajío to keep a disrupted project on track. At the Week-4 review, the project was slightly behind schedule (SPI 0.93) and over budget (CPI 0.957) because of a staff sick-leave delay and a 4% equipment cost increase, but processing the five reported events through formal change control — extending the highest-volume activity, crashing the critical path, and applying the contingency reserve — enabled recovery of the schedule even under management's request to advance go-live by one week. The project ultimately achieved its strategic objective: consolidating the three branches onto a centralized cloud platform, eliminating the 53,208 duplicated records to reach 0% redundancy, achieving sub-2-second query response, and delivering by July 31, 2026.

Strategic Impact.

Beyond the immediate project deliverables, the Medical Information System positions Centro Médico del Bajío for lasting operational improvements. By adopting a cloud SaaS model, the organization gains enterprise-grade data management without the capital cost and operational burden of on-premise infrastructure. Real-time access to unified patient records across the three branches improves clinical decision-making, reduces duplicate testing, and strengthens regulatory compliance. The project also lays the technical foundation for future capabilities — telemedicine integration, clinical analytics dashboards, and interoperability with Mexico's national health information systems — converting what began as a data-consolidation problem into a strategic digital-transformation asset.

Lessons Learned.

Execution confirmed the value of disciplined monitoring and control. Earned Value Analysis surfaced the schedule and cost slip early enough to act, while variance analysis correctly attributed it to a sick-leave absence, record-volume underestimation, and equipment inflation rather than systemic overrun. The experience reinforced four practices for future projects:

- Assign a backup owner for every critical-path activity to absorb absences without cascading impact (C1 was delayed 2 days because no one else could start the activity).
- Estimate digitization with a per-record parametric model instead of a flat duration (C6 required 8 days at the measured throughput of 4,350 records/day, not the originally estimated 5 days).
- Size and ring-fence a contingency reserve to absorb price volatility and sponsor-driven scope changes (the 4% equipment increase and the schedule recovery both relied on the 10% contingency).
- Standardize Earned Value templates so the schedule and cost variance formulas are not confused (the SV/CV swap on the whiteboard could have led to incorrect corrective actions if not caught).

Future Recommendations.

- Establish a post-go-live benefits-realization review at 3 and 6 months to measure the actual reduction in duplicate records, query response times, and operational cost savings against the baseline.
- Implement a clinical analytics layer on the centralized cloud database to provide dashboards for patient outcomes, resource utilization, and branch performance comparisons.

- Formalize a cybersecurity review cycle for the cloud platform (access control audit, encryption validation, and vendor SLA compliance checks).
- Define an onboarding standard so that new patient records, branches, or clinical document types are provisioned through the centralized platform by default, preserving 0% redundancy.
- Evaluate telemedicine integration using the cloud API to extend the platform's value to remote consultations across the three branches.

4. Bibliography

References follow the APA 7th edition style.

Heagney, J. (2011). *Fundamentals of Project Management* (4th ed.). American Management Association.

Project Management Institute. (2021). *A Guide to the Project Management Body of Knowledge (PMBOK Guide)* (7th ed.). Project Management Institute.

Watt, A. (2014). *Project Management*. BCcampus Open Textbook Project. https://opentextbc.ca/projectmanagement/

Universidad Politécnica de Aguascalientes. (2026). *Guidelines for Managing Information Technology Project Plans*. Internal Academic Document.

The Open University of Hong Kong. (2016). *Project Management*. Open Textbooks for Hong Kong. http://open.bccampus.ca/